

Assignment: Understanding Python Data Structures Through Real-Life Examples

Objective: Understand List, Tuple, Set, and Dictionary using simple real-life scenarios.

Question 1: Shopping List (List)

Create a list containing 10 grocery items.

1. Add two new items.
2. Remove one item.
3. Display the first and last item.
4. Sort the list alphabetically.
5. Count the total number of items.

Question 2: Mobile Contacts (Tuple)

Create a tuple containing 5 mobile numbers.

1. Display all numbers.
2. Find the length of the tuple.
3. Access the third mobile number.
4. Check whether a given number exists in the tuple.

Observation: Why can't we modify a tuple after creation?

Question 3: Unique Visitors (Set)

Store visitor IDs: 101, 102, 103, 101, 105, 103, 106 in a set.

1. Display unique visitor IDs.
2. Add a new visitor.
3. Remove a visitor.
4. Find the total number of unique visitors.

Observation: Why are duplicate values removed automatically?

Question 4: Student Information (Dictionary)

Create a dictionary storing Name, Age, Course, and City.

1. Display student details.
2. Add Phone Number.
3. Update City.
4. Delete Age.
5. Display all keys and values.

Question 5: Comparison Activity

Create one List, Tuple, Set, and Dictionary containing your personal information. Compare Ordered, Mutable, and Duplicate Allowed properties.

Question 6: Real-Life Mapping

Identify one real-life example for each:

List, Tuple, Set, and Dictionary.

Explain each example in 3–4 lines.

Question 7: Mini Challenge

Create a program that stores information about 5 friends using a dictionary. Display Friend Name, Favorite Color, and Favorite Food. Allow the user to search for a friend's details.

Thought Questions (No Coding)

1. Why do we need different data structures?
2. Can one data structure solve all problems? Why?
3. Which data structure is most useful in daily life and why?